Varunprasaath R. Selvaraju, Ph.D.

(+1) 678-207-9914 | varunprasaath@gmail.com | linkedin.com/in/varunprasaath-rs |

PROFESSIONAL SUMMARY

PhD chemist/materials scientist with analytical method-development, chromatography-adjacent LC-MS/GC-MS exposure, electrochemical characterization, DOE/SPC, and technical documentation experience. Builds reproducible analytical workflows and structure-property datasets for process-control and semiconductor-adjacent chemical systems.

CORE COMPETENCIES

- Analytical Method Development
- LC/GC-MS-Adjacent Chromatography
- Process Fluids / Process Control
- DOE / QbD / SPC
- Method Validation Mindset
- Electrochemistry & Cyclic Voltammetry
- Spectroscopy & Thermal Analysis
- Technical Reports / GLP-Style Documentation

WORK EXPERIENCE

University of Colorado Boulder
July 2025 - Present

Postdoctoral Researcher

- Lead DOE-guided chemical process-development experiments for bio-based polymer systems, defining analytical readouts, process windows, and reproducible specifications for scale-up.
- Use FT-IR, thermal analysis, spectroscopy, and statistical process-control concepts to connect process parameters with material quality and method repeatability.
- Document experimental methods, calibration expectations, and technical recommendations in SOP-ready formats for repeatable execution by cross-functional lab teams.

University of Colorado Boulder
July 2021 - July 2025

Graduate Researcher

- Designed and characterized 15+ electrochemical material variants using cyclic voltammetry, spectroelectrochemistry, UV-Vis-NIR, FT-IR, NMR, GPC, TGA, DSC, and LC/GC-MS-adjacent analytical interpretation to quantify structure-property-performance relationships.
- Applied electrochemical stress testing and degradation analysis to identify redox instability, oxidative degradation, and interface failure modes, then recommended design/process mitigations.
- Built rigorous experimental documentation and analytical datasets supporting method comparison, reproducibility, technical reports, and publication-quality presentations.

Georgia Institute of Technology
Aug 2019 - July 2021

Graduate Researcher

- Co-led AFOSR-funded process development for molecular thin-film coatings, developing surface-preparation and CVD process controls to achieve target optical/material performance.
- Connected processing conditions with film morphology and device behavior through spectroscopy, thin-film characterization, and iterative experimental design.

ETH Zurich
May 2017 - July 2017

Research Assistant

- Optimized polymer thin-film deposition using DOE strategies and in-situ QCM-D process monitoring, characterizing film-growth kinetics to improve coating reproducibility.

University of Colorado Boulder
2021 - 2025

Laboratory Operations Manager

- Managed operations for a 15-member process-development laboratory, implementing equipment tracking, calibration routines, SOPs, and safety protocols that improved instrument uptime by 30%.
- Trained researchers on instrument workflows, experimental documentation, and safe handling while coordinating multiple concurrent materials-development projects.

Georgia Tech
2019-2021

Teaching Assistant, Organic Chemistry I & Lab

- Mentored 120+ undergraduates; developed supplemental materials improving exam performance by 15%.

KEY PROJECTS

Electrochemical Materials Method Development
Electrochemistry

Designed and evaluated functional electrochromic materials using cyclic voltammetry, spectroelectrochemistry, and stability testing to connect redox behavior with performance and degradation.

Cyclic voltammetry, spectroelectrochemistry, UV-Vis-NIR, FT-IR, DOE, root-cause analysis

Process-Controlled Polymer Systems
Process Development

Led DOE-guided process optimization for polymer systems, defining critical parameters, analytical readouts, and reproducible process specifications.

DOE, SPC, FT-IR, thermal analysis, SOPs

Thin-Film Coating Process Development
Semiconductor-adjacent Materials

Developed CVD/surface-preparation strategies for molecular coatings, translating process variables into film morphology and optical/material performance.

CVD, QCM-D, spectroscopy, surface preparation

EDUCATION

Ph.D. in Chemistry - University of Colorado Boulder 2025

Dissertation: Modular Design and Electrochemical Properties of Xanthene-Based Electrochromic Materials
Advisor: Prof. Seth Marder
(Transferred from Georgia Institute of Technology)

B.S. (Research), Major: Chemistry; Minor: Materials Science - Indian Institute of Science (IISc) 2019

INSPIRE Scholarship for Higher Education (Top 1% nationally)

CERTIFICATIONS

Statistical Process Control (SPC) - Coursera, 2026

TECHNICAL SKILLS

Analytical Chemistry: LC-MS and GC-MS exposure, cyclic voltammetry, spectroelectrochemistry, UV-Vis-NIR, FT-IR, TGA, DSC, NMR, GPC, QCM-D; chromatography-method principles without overclaiming HPLC ownership
Method & Process Development: DOE, QbD/SPC mindset, analytical method comparison, validation documentation, process windows, reproducibility studies, SOPs, technical reports
Materials & Process Fluids: Organic/organometallic synthesis, polymer systems, thin films/coatings, electrochemical materials, semiconductor process-control adjacency
Data & Communication: Root-cause analysis, degradation/stability testing, Excel, Python/basic data analysis, JMP/basic, conference presentations and peer-reviewed publications

JOURNAL PUBLICATIONS

1. Ramasamy Selvaraju, V.; Venta, R.; Rajapaksha, I.; Zhang, J.; Barlow, S.; Salman, S.; Scott, C.; Marder, S. "Narrow-Band Electrochromism of a Ferrocene-Substituted Rhodamine B." Chemistry of Materials, 2025, 37, 5558-5568.

2. Mandal, J.; Ramasamy Selvaraju, V.; Yan, W.; Divandari, M.; Spencer, N. D.; Dubner, M. "In Situ Monitoring of SI-ATRP Throughout Multiple Reinitiations Under Flow by Means of a Quartz Crystal Microbalance." RSC Advances, 2018, 8, 20048-20055.
3. Ramasamy Selvaraju, V.; Venta, R.; Rajapaksha, I.; Zhang, J.; Barlow, S.; Salman, S.; Scott, C.; Marder, S. "Beyond Rhodamine B: Investigating Alternative Xanthene Cores for Modular Electrochromic Systems." (Manuscript under preparation).

CONFERENCE PRESENTATIONS

Modular Design and Electrochemical Properties of Xanthene-Based Electrochromic Materials, ACS Fall National Meeting, Denver, CO, August 2024.
Electrochemical and Spectroscopic Properties of Substituted Rhodamine Derivatives, Gerischer-Lewerenz Electrochemistry Symposium, Fort Collins, CO, August 2024.
Decoupling Redox and Color for High-Contrast Electrochromics, Excited State Processes Conference, Santa Fe, NM, June 2023.
Electrochromism: Decoupling Color and Redox in Rhodamine-Based Systems, Rocky Mountain Solid State Chemistry Workshop, Boulder, CO, January 2023.

HONORS & AWARDS

Plastic Sustainability Innovation in Material Award | INSPIRE Scholarship (Top 1%, India, 2015) | 97.12 Percentile - IIT JEE Mains (2015) | All India Rank 144 - NEST (2014)